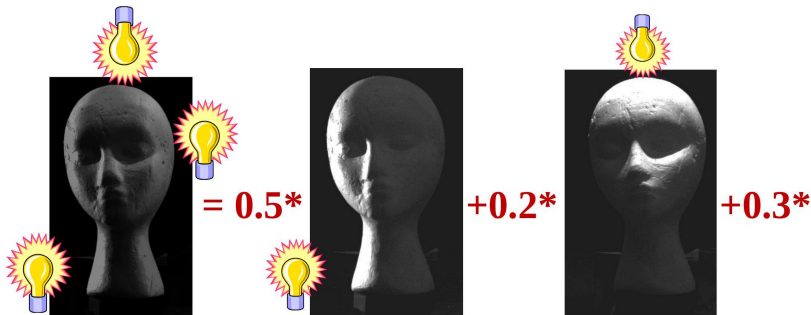


Lambertian model of reflectance II: harmonic analysis

Illumination cone

- ▶ What is the set of images of an object under different lighting, with any number of sources?
- ▶ Images are additive and non-negative
- ▶ This set, therefore, forms a convex cone in $\mathbb{R}^p$, p number of pixels (Belhumeur & Kriegman)

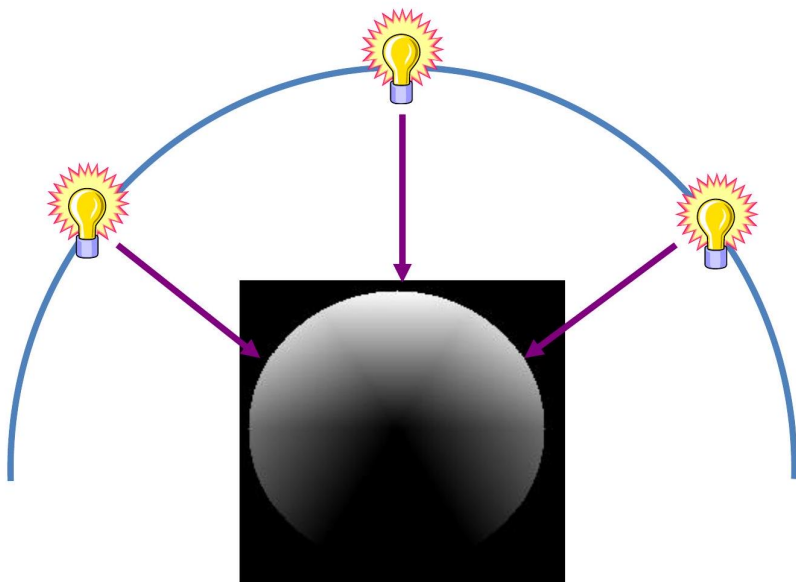


Illumination cone

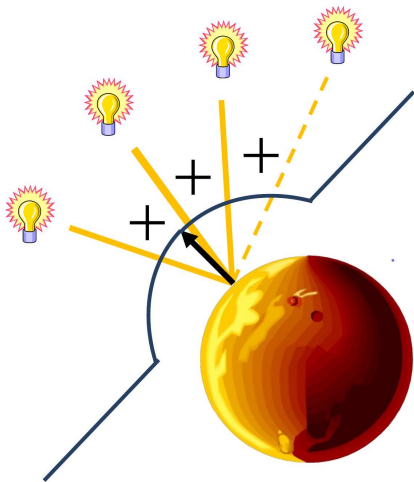
- ▶ Cone characterization is generic, holds also with specularities, shadows and interreflections
- ▶ Unfortunately, representing the cone is complicated (infinite degrees of freedom)
- ▶ Cone is "thin" for Lambertian objects; indeed the illumination cone of many objects can be represented with few PCA vectors (Yuille et al.)

Lambertian reflectance is smooth

(Basri & Jacobs; Ramamoorthi & Hanrahan)

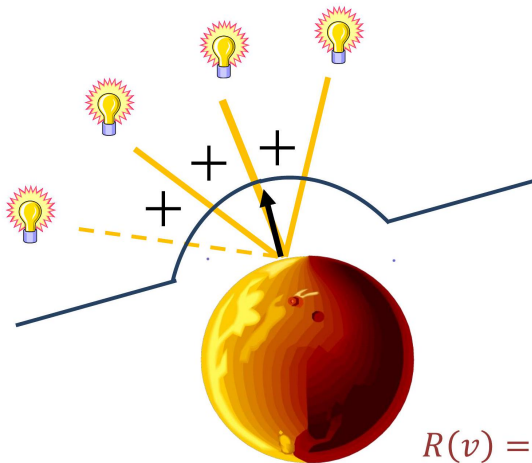


Reflectance obtained with convolution



$$R(v) = \int_{S^2} k(u, v) l(u) du$$

Reflectance obtained with convolution



$$R(v) = \int_{S^2} k(u, v) l(u) du$$

Spherical harmonics

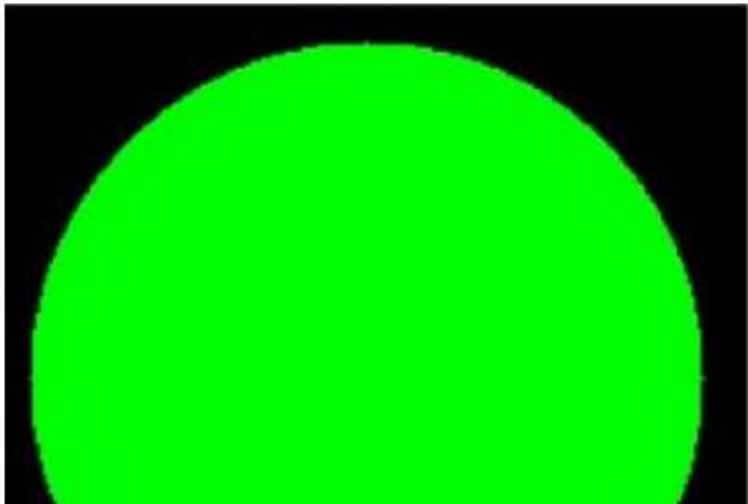
$$Y_{nm}(\theta, \varphi) = \sqrt{\frac{(2n+1)}{4\pi} \frac{(n-|m|)!}{(n+|m|)!}} P_{nm}(\cos \theta) e^{im\varphi}$$

$$p_{nm}(z) = \frac{(1-z^2)^{m/2}}{2^n n!} \frac{d^{n+m}}{dz^{n+m}} (z^2 - 1)^n$$

- ▶ Orthonormal basis for functions on the sphere
- ▶ n 'th order harmonics have $2n + 1$ components
- ▶ Rotation = phase shift (same n , different m)
- ▶ In space coordinates: polynomials of degree n
- ▶ Funk-Hecke convolution theorem

Spherical harmonics

$$X^2 + Y^2 + Z^2 = 1$$



Harmonic approximation

- ▶ Lighting, in terms of harmonics

$$\ell(\theta, \phi) = \sum_{n=0}^{\infty} \sum_{m=-n}^n l_{nm} Y_{nm}(\theta, \phi)$$

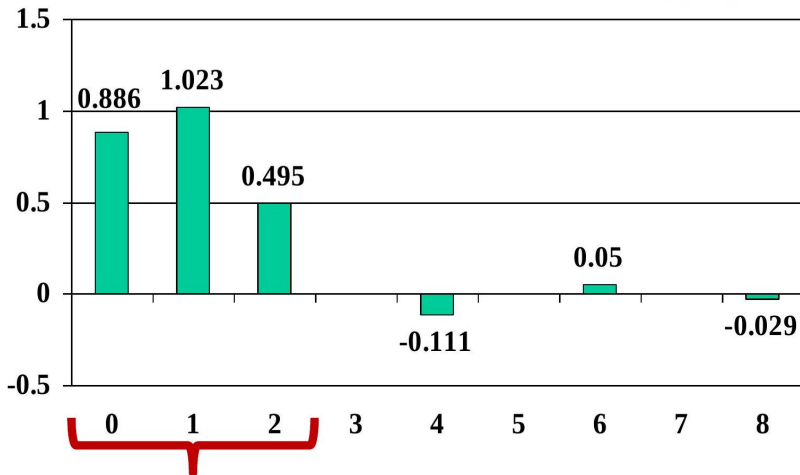
- ▶ Reflectance

$$r(\theta, \phi) = k * \ell \approx \sum_{n=0}^2 \sum_{m=-n}^n k_n l_{nm} Y_{nm}(\theta, \phi)$$

- ▶ Approximation accuracy, 99.2%
(Basri & Jacobs; Ramamoorthi & Hanrahan)

Harmonic transform of kernel


$$k(\theta) = \max(\cos \theta, 0) = \sum_{n=0}^{\infty} k_n Y_{n0}$$

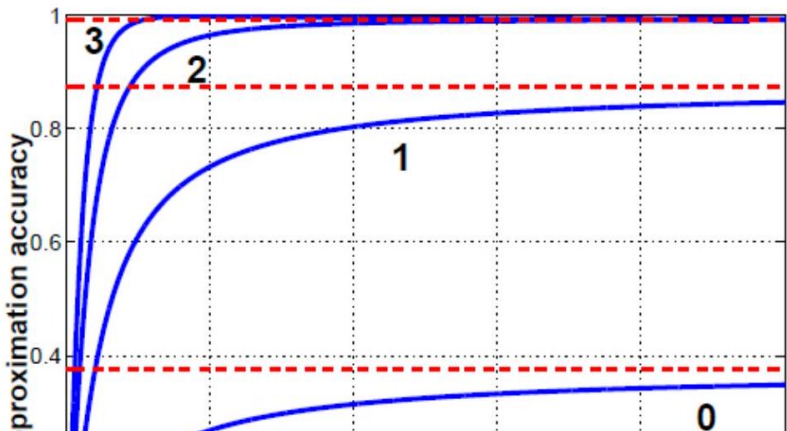


Subspace approximation

- ▶ Up to 2nd order:
 - ▶ 9 basis images
 - ▶ Accuracy: 99.2%
- ▶ Up to 1st order:
 - 4 basis images: ambient + point source
 - ▶ Accuracy: 87.5%
- ▶ In practice, due to self occlusions $\sim 98\%$ can be achieved with just 6 basis images (Ramamoorthi)

Scope

- ▶ Harmonic representations handle convex, lambertian objects with multiple light sources (including attached shadows)
- ▶ Harmonic representations do not model cast shadows and inter-reflections
- ▶ Accuracy is maintained for fairly close light sources
- ▶ Representing specular objects may require a very large basis



Applications

- ▶ We can use this theory to predict novel appearances under new lighting
- ▶ Harmonic lighting theory has led to applications in
- ▶ Face recognition
- ▶ Photometric stereo
- ▶ 3D reconstruction with prior
- ▶ Motion analysis

"Harmonic faces"



Positive values



Negative values



ρ Albedo

n Surface normal

$n = (n_x, n_y, n_z)$

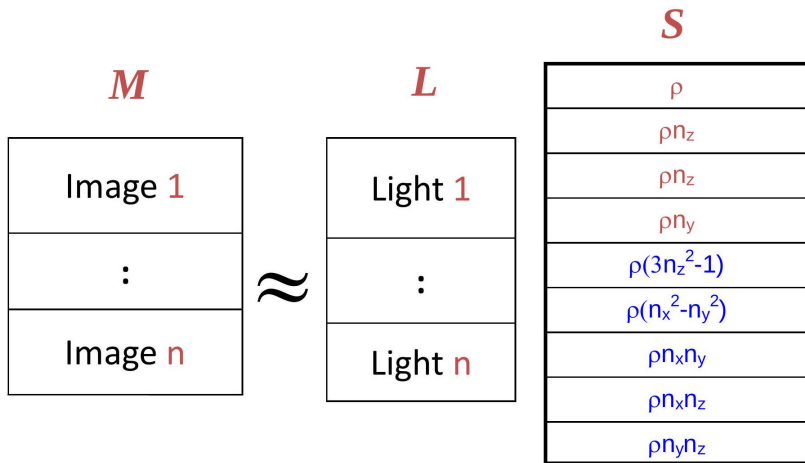


Non-negative light

- ▶ We can enforce in addition that light is nonnegative, by projecting the illumination cone onto the harmonic space
- ▶ Closed-form constraints for 1st order approximation
- ▶ Sampling method, or Toeplitz matrix (Shirdhonkar & Jacobs) for higher orders

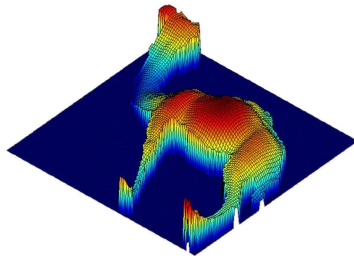
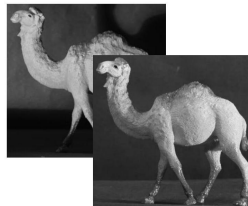
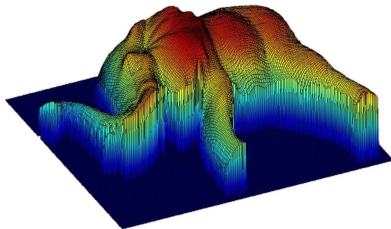
Photometric stereo

(Basri, Jacobs & Kemelmacher)



SVD recovers L and S up to an $(r \times r)$ ambiguity

Photometric stereo



Reconstruction with a prior

- ▶ Given just one image SFS is impractical
- ▶ Reconstruction is possible when a prior is available
- ▶ Energy

$$\min_{I, \rho, Z} \iint_{\Omega} D + S$$

- ▶ Data term

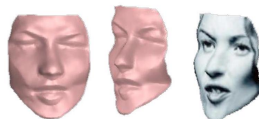
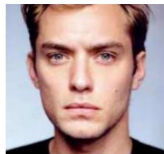
$$D = \left(I - \rho l^T Y(\hat{n}) \right)^2$$

- ▶ Regularization

$$S = \lambda_1 (\Delta (Z - Z_{\text{prior}}))^2 + \lambda_2 (\Delta (\rho - \rho_{\text{prior}}))^2$$

- ▶ Solve as a linear PDE

Reconstruction with a prior



More...

